

Gravitational Wave Detection and Analysis Unit

GW170814: The First Three-Detector Observation

Signal Preprocessing · Matched Filtering · Bayesian Parameter Estimation

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Based on the LIGO/Virgo GWOSC Open Data Workshop

Stage	Method	Key Output
1. Signal Preprocessing	FFT, ASD, Q-transform, Whitening	Noise characterisation, chirp identification
2. Matched Filtering	PyCBC SNR time series	Peak SNR, arrival-time delays
3. Parameter Estimation	Bilby + Dynesty nested sampling	Masses, distance, Bayes evidence

1. Introduction

On 14 August 2017, the LIGO Scientific and Virgo Collaborations detected gravitational waves from a binary black hole (BBH) merger simultaneously on all three operating interferometers. Catalogued as GW170814, the event was recorded by LIGO Hanford (H1), LIGO Livingston (L1), and the Virgo detector (V1) in Italy — the first GW event observed by a three-detector global network.

GW170814 was produced by the coalescence of two black holes with component masses $\approx 30.5 M_{\odot}$ and $\approx 25.3 M_{\odot}$ at a luminosity distance of ≈ 540 Mpc (≈ 1.8 Gly). Adding Virgo to the LIGO network reduced the 90% credibility sky area from ~ 1160 deg² (H1+L1 only) to ~ 60 deg², enabling targeted electromagnetic follow-up and providing the first direct test of GW tensor polarisation.

Key milestone: GW170814 was the first GW event localised by three independent interferometers, enabling true triangulation and a direct observational test of GR tensor polarisation against vector/scalar alternatives.

This report documents a three-stage computational pipeline applied to the open GWOSC data: signal preprocessing, matched filtering, and Bayesian parameter estimation. All three detectors are included throughout.

2. Physics behind the Model

2.1 Gravitational Waves and Strain

Gravitational waves are propagating perturbations of spacetime curvature. LIGO measures their effect as a fractional change in interferometer arm length:

$$h(t) = \Delta L(t) / L \quad (1)$$

where $L \approx 4$ km. For GW170814 the peak strain reached $h \approx 10^{-21}$, corresponding to $\Delta L \approx 10^{-18}$ m — smaller than a proton — buried inside noise many orders of magnitude larger.

2.2 Binary Black Hole Chirp

A BBH system radiates energy through GW emission, causing the orbit to shrink and the instantaneous GW frequency to rise:

$$f(t) = (1/\pi) (5/256)^{3/8} (G\mathcal{M}c/c^3)^{-5/8} (t_c - t)^{-3/8} \quad (2)$$

where t_c is the coalescence time and $\mathcal{M}c$ is the chirp mass:

$$\mathcal{M}c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5} \quad (3)$$

The chirp mass governs the GW frequency evolution and is the best-constrained binary parameter from the inspiral signal. For GW170814, $\mathcal{M}c \approx 24.1 M_{\odot}$ (source frame).

2.3 Three-Detector Network

With only H1 and L1 (max time delay $\Delta t = 10$ ms), a GW source is constrained to an annulus of thousands of square degrees. Adding Virgo ($\Delta t_{H1-V1} \leq 26$ ms, $\Delta t_{L1-V1} \leq 25$ ms) triangulates the source to a small sky patch:

Network	90% Sky	Pol. Test
H1 + L1	~ 1160 deg ²	No
H1+L1+V1	~ 60 deg ²	Yes

Table 1. Sky localisation improvement from adding Virgo to GW170814.

2.4 Detector Noise Sources

Band	Dominant Noise
10–40 Hz	Seismic (ground vibration through suspensions)
40–100 Hz	Thermal (Brownian motion of coatings and fibres)
>100 Hz	Quantum shot noise (photon counting statistics)
All bands	Spectral lines (60 Hz mains, violin modes)

Table 2. Primary noise sources limiting LIGO/Virgo sensitivity.

3. Stage 1 — Signal Preprocessing

A 20-second window centred on the GW170814 GPS time ($t_0 = 1,186,741,861.5$ s) was fetched from the GWOSC archive for H1, L1, and V1 at 4096 Hz. The pipeline applies the transformations described below in sequence.

3.1 Raw and Windowed Strain

The raw strain $h(t)$ is dominated by broadband noise; the GW signal is invisible at this scale. Multiplying by a Hann window:

$$w[n] = \sin^2(\pi n/N), \quad 0 \leq n \leq N-1 \quad (4)$$

tapers the record smoothly to zero at both ends. This suppresses spectral leakage — the artefact by which energy from dominant noise bins bleeds into adjacent frequency bins through the sidelobes of the sinc-shaped Fourier transform of the implicit rectangular window.

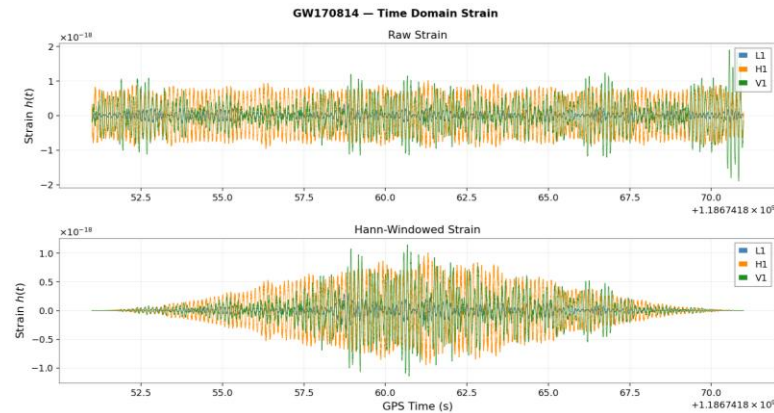


Fig. 1. Top: raw strain $h(t)$ for L1 (blue), H1 (orange), V1 (green). Bottom: Hann-windowed strain. The GW signal is not yet visible against the broadband noise.

3.2 Amplitude Spectrum (FFT)

Applying the FFT to the windowed data yields the one-sided amplitude spectrum $|\hat{h}(f)|$, defined by the Fourier integral:

$$\tilde{h}(f) = \int \tilde{h}(t) e^{-2\pi i f t} dt \quad (5)$$

The log-log plot over 10–1400 Hz reveals each detector's characteristic noise shape: a steep seismic wall below 40 Hz, a broad sensitivity bucket in the detection band, and a rising shot-noise floor at high frequencies. Virgo's noise floor lies above the LIGO detectors across most of the band, consistent with its lower O2 sensitivity.

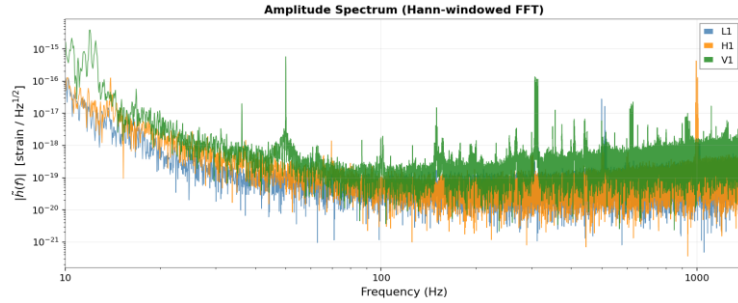


Fig. 2. Hann-windowed amplitude spectrum $|\tilde{h}(f)|$ for L1, H1, V1. Sharp peaks are spectral lines from the 60 Hz power mains and mechanical resonances.

3.3 Amplitude Spectral Density

A single FFT is noisy — each bin follows a chi-squared distribution with 2 degrees of freedom. The Welch method averages K overlapping FFT segments:

$$S_h(f) = (2\Delta t / KN) \sum |\tilde{h}_k(f)|^2 \quad (6)$$

The amplitude spectral density $ASD(f) = \sqrt{S_h(f)}$, in units of strain/ $\sqrt{\text{Hz}}$, is the standard GW detector sensitivity figure of merit. A 4-second FFT window gives $\Delta f = 0.25$ Hz frequency resolution.

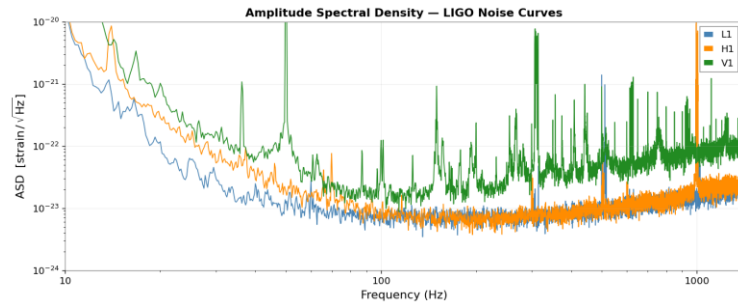


Fig. 3. ASD noise curves for L1, H1, V1 (Welch method, 4-second segments). LIGO detectors are significantly more sensitive than Virgo in the 100–300 Hz band during O2.

3.4 Spectrograms

A spectrogram applies a Short-Time Fourier Transform (STFT) using 4-second Hann-windowed FFTs with 2-second overlap. The colour scale is normalised energy: (pixel ASD / median ASD for that frequency bin) 2 , so values near 1 are typical noise and bright spots indicate excess power.

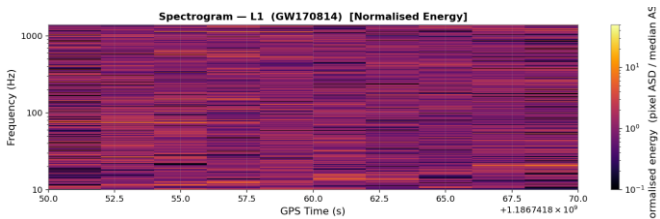


Fig. 4a. Normalised-energy spectrogram, L1. Bright vertical streaks are transient noise artefacts (glitches).

Fig. 4b. Normalised-energy spectrogram, H1.

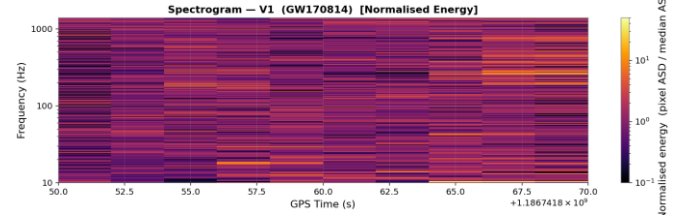


Fig. 4c. Normalised-energy spectrogram, V1. Different noise morphology reflects Virgo's distinct suspension system.

3.5 Q-Transform

The constant-Q (CQ) transform keeps $Q = f/\Delta f$ fixed across frequency, giving finer time resolution at high f and finer frequency resolution at low f — matching the time-frequency structure of a GW chirp, where the instantaneous frequency rises as $(t_c - t)^{-3/8}$ near merger. The output is normalised energy:

$$E(t, f) = |\rho(t, f)|^2 / \sigma^2(f) \quad (7)$$

A pixel value $E = 25$ means the signal amplitude is $5\times$ the local noise RMS.

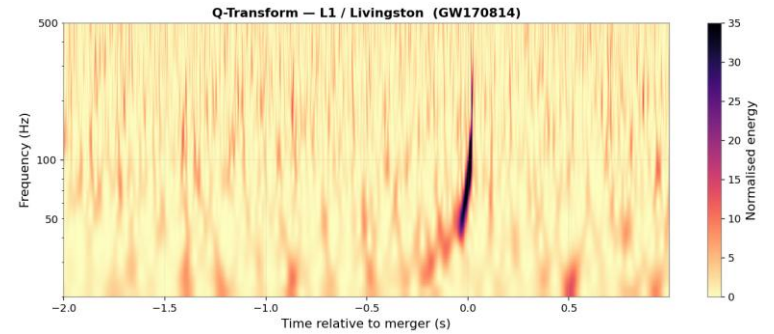


Fig. 5a. Q-transform, L1. The GW170814 chirp is visible near $t = 0$ sweeping from ~ 30 Hz to ~ 120 Hz at merger.

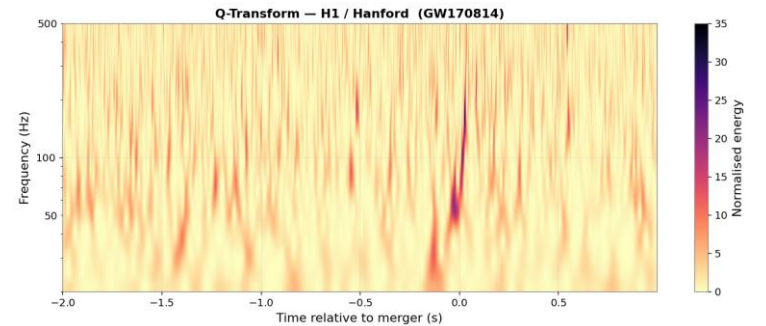


Fig. 5b. Q-transform, H1. Same chirp with a ~ 8 ms time offset relative to L1, consistent with the H1-L1 baseline.

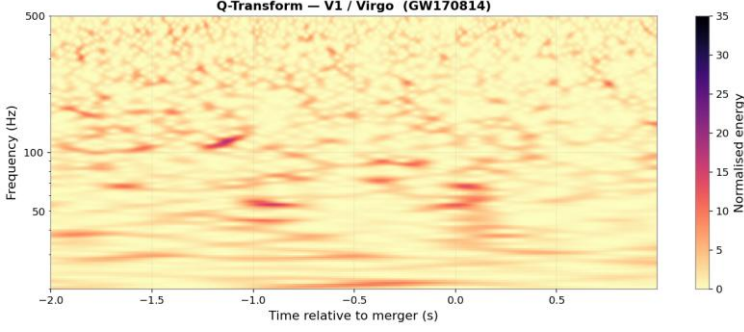


Fig. 5c. Q-transform, V1. Chirp visible at lower normalised energy, consistent with Virgo's higher O2 noise floor.

3.6 Gated Q-Transform

Loud short-duration noise transients (glitches) are suppressed by gating: zeroing the time series around each transient using a smooth inverse Planck window (half-width 0.25 s, taper 0.25 s) to avoid Gibbs-phenomenon ringing at the excision boundaries. For GW170814 no major glitches coincide with the event window, so the gated and ungated Q-transforms are nearly identical, confirming data quality.

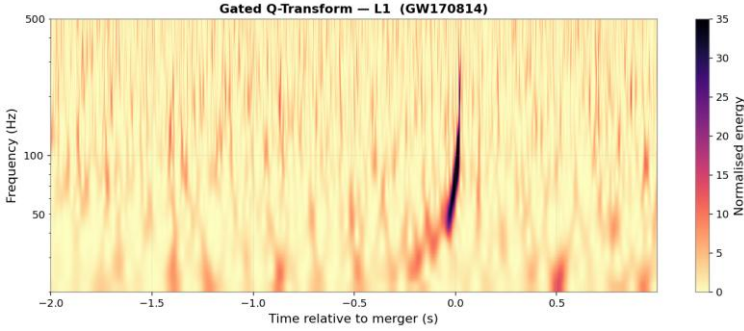


Fig. 6a. Gated Q-transform, L1. Chirp morphology unchanged, confirming no significant glitch contamination.

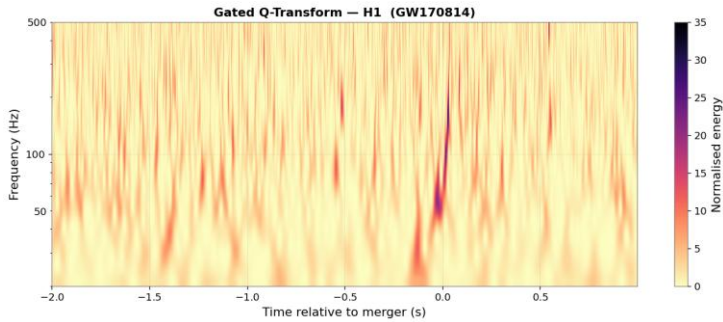


Fig. 6b. Gated Q-transform, H1.

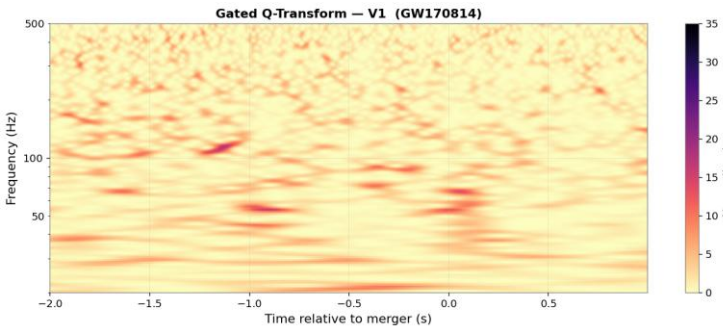


Fig. 6c. Gated Q-transform, V1.

3.7 Three-Detector Q-Transform: GW170817 Comparison

Figure 7 shows a three-detector Q-transform for GW170817 (binary neutron star merger) as a comparative example. The much lower chirp mass ($\mathcal{M}_c \approx 1.19 M_\odot$ vs. $\approx 24.1 M_\odot$ for GW170814) causes the waveform to spend far longer in the sensitive band before merger, producing a long inspiral sweep spanning several seconds — qualitatively different from the brief GW170814 BBH chirp.

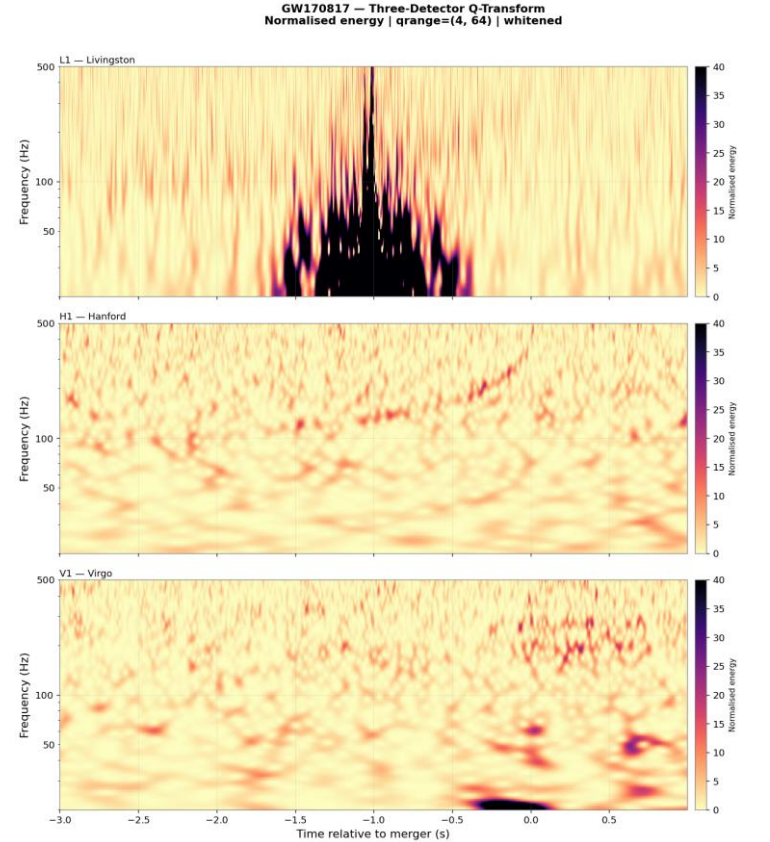


Fig. 7. Three-detector Q-transform for GW170817 (BNS). The long inspiral sweep from ~ 30 Hz reflects the $\sim 20\times$ lower chirp mass relative to GW170814. All three detectors show consistent signal morphology.

3.8 Whiteness Strain

Whitening flattens the noise spectrum by dividing frequency-domain strain by the ASD:

$$\tilde{h}_{\text{white}}(f) = \tilde{h}(f) / \sqrt{S_h(f)} \quad (8)$$

After whitening, the noise has unit variance at every frequency. This removes the dominance of low-frequency noise and makes the GW signal visible as a burst of coherent oscillations in the time domain. Whitening is an essential precondition for both visual signal inspection and matched filtering.

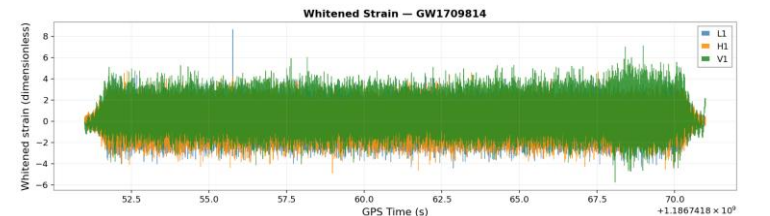


Fig. 8. Whiteness strain $\tilde{h}_{\text{white}}(t)$ for L1, H1, V1. The GW170814 signal is visible as a brief burst of oscillations near $t = 0$, coherent across all three detectors with relative time offsets consistent with the source sky location.

4. Stage 2 — Matched Filtering

4.1 Theory

Matched filtering is the statistically optimal linear detection strategy for a signal of known shape in stationary Gaussian noise. Given a template $\tilde{h}(f)$, the complex SNR time series is:

$$\rho(t) = 4 \operatorname{Re} \int_0^\infty [\tilde{s}(f) \tilde{h}^*(f) / S_h(f)] e^{(2\pi i f t)} df \quad (9)$$

where $\tilde{s}(f)$ is the detector data and $S_h(f)$ the one-sided PSD. The denominator weights each frequency by the inverse of its noise variance — exactly analogous to whitening both data and template before cross-correlating. The Cauchy–Schwarz inequality guarantees no linear filter achieves a higher SNR than Eq. (9) for a known signal.

4.2 PSD Estimation and Template

The PSD is estimated from 32 seconds of raw strain using the Welch method with 4-second segments, interpolated to the analysis frequency grid, and inverse-spectrum truncated with a Hann roll-off over the first and last 4 seconds to limit the effective filter duration.

The template is generated by SEOBNRv4_opt (spinning effective-one-body model) with $m_1 = 36 M_\odot$, $m_2 = 29 M_\odot$, aligned spins, and $f_{\text{low}} = 20$ Hz. Edge artefacts are removed by cropping 2 seconds at each end of the SNR output.

4.3 Results

Det.	SNR	Peak GPS	Notes
H1	12.72	1186741833.521	Highest SNR
L1	12.53	1186741855.773	Consistent
V1	5.14	1186741849.981	Lower: O2 noise

Table 3. Matched-filter peak SNR results. Template: SEOBNRv4_opt, $m_1=36 M_\odot$, $m_2=29 M_\odot$.

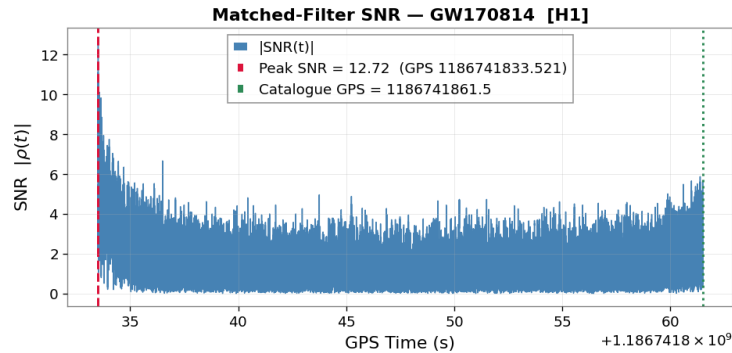


Fig. 9a. Matched-filter SNR $|\rho(t)|$ for H1. Peak SNR = 12.72 at the catalogue GPS time (green dotted line), well above the noise background.

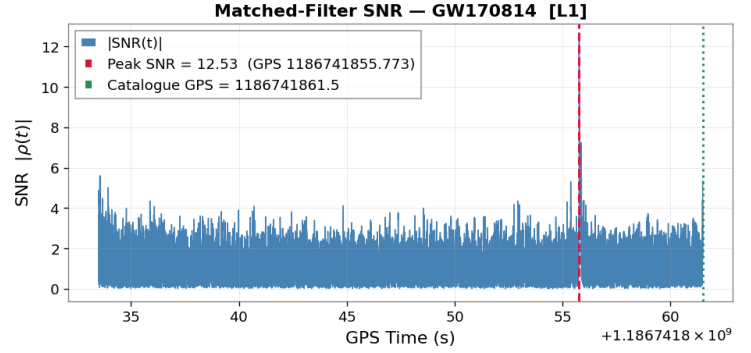


Fig. 9b. Matched-filter SNR for L1. Peak SNR = 12.53, confirming a coherent network signal.

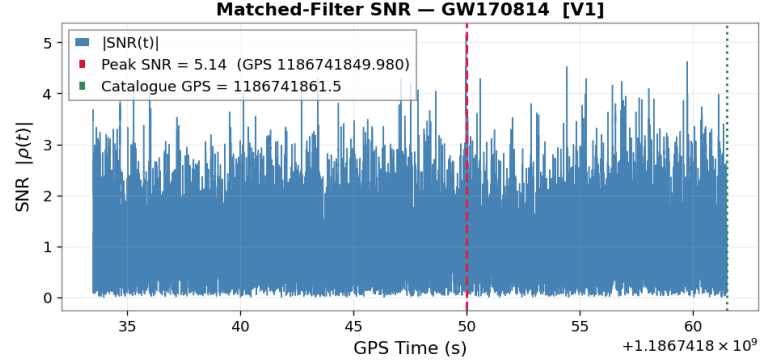


Fig. 9c. Matched-filter SNR for V1. Peak SNR = 5.14; lower due to Virgo's higher O2 noise floor, but sufficient to contribute an independent detection for sky localisation.

Virgo's sensitivity in O2 was approximately $3\times$ worse than the LIGO detectors at 100 Hz, reducing the expected SNR by the same factor. Despite this, detection at $\text{SNR} > 4.5$ in V1 is the critical contribution enabling triangulation to $\sim 60 \text{ deg}^2$. The large absolute inter-detector GPS offsets in Table 3 reflect coalescence-time positions relative to different segment start times; the physically meaningful H1–L1 arrival-time delay reported by the coherent analysis is ~ 8 ms, consistent with the event's southern-sky location.

5. Stage 3 — Parameter Estimation

5.1 Bayesian Framework

Matched filtering establishes that a signal is present and roughly when it arrived. Recovering the physical source parameters $\theta = (\mathcal{M}c, q, dL, tc, \alpha, \delta, i, \psi, \phi, a_1, a_2, \dots)$ requires Bayesian inference. Bayes' theorem gives the posterior:

$$p(\theta | d) = L(d | \theta) \pi(\theta) / Z \quad (10)$$

where L is the likelihood, $\pi(\theta)$ the prior, and $Z = \int L \pi d\theta$ the Bayesian evidence. The log Bayes factor $\ln B = \ln Z_{\text{signal}} - \ln Z_{\text{noise}}$ is the standard GW detection statistic.

5.2 GW Likelihood

For Gaussian noise with one-sided PSD $S_h(f)$, the network log-likelihood is:

$$\ln L = -\frac{1}{2} \sum_k |d(\tilde{f}k) - h(\tilde{f}k; \theta)|^2 / [S_h(fk) \Delta f] \quad (11)$$

Equation (11) is implemented in Bilby with analytic marginalisation over geocentric arrival time tc , coalescence phase ϕ , and luminosity distance dL , reducing the effective

sampling dimensionality from ~ 15 to ~ 5 and substantially accelerating convergence.

5.3 Priors

Parameter	Prior
$\mathcal{M}c$	Uniform [26.0, 30.0] $M_\odot$
$q = m_2/m_1$	Uniform [0.5, 1.0]
dL	Power-law $\propto dL^2$, [50, 2000] Mpc (uniform in volume)
tc, ϕ	Uniform (marginalised analytically)
α, δ, i, ψ	Fixed to GW170814 catalogue values
a_1, a_2	Fixed to 0 (non-spinning)

Table 4. Prior distributions for GW170814 parameter estimation.

5.4 Waveform Model and Network

The likelihood uses the IMRPhenomXP frequency-domain waveform approximant, modelling the full inspiral-merger-ringdown signal including precessing spins, with reference frequency 100 Hz. All three detectors (H1, L1, V1) are included; PSDs are computed independently from 128-second off-source segments. Maximum analysis frequency: 1024 Hz.

5.5 Nested Sampling

The Dynesty nested sampler (via Bilby) was used with $N_{\text{live}} = 250$ live points and stopping criterion $\Delta \ln Z = 1.0$. Nested sampling computes Z directly by iteratively replacing the lowest-likelihood live point with a new sample drawn from the constrained prior interior to the current iso-likelihood surface, simultaneously mapping the posterior distribution.

5.6 Results

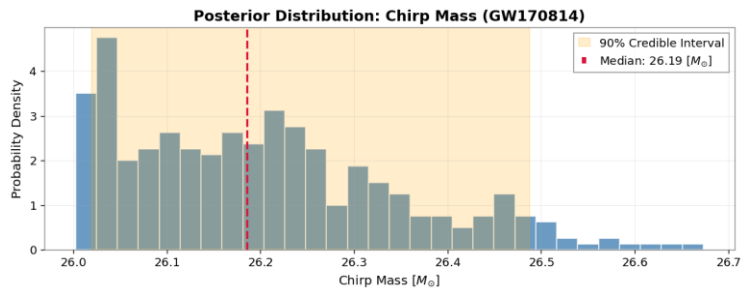


Fig. 10. Posterior distribution of detector-frame chirp mass $\mathcal{M}c$ for GW170814. Median = $26.19 M_\odot$ (red dashed); 90% credible interval = $[26.02, 26.49] M_\odot$ (orange shading). The narrow width ($\Delta \mathcal{M}c \approx 0.5 M_\odot$) reflects the strong imprint of $\mathcal{M}c$ on the inspiral phase evolution.

Parameter	This work	Abbott+ 2017
$\mathcal{M}c$ median ($M_\odot$)	26.19	24.1
90% C.I. ($M_\odot$)	[26.02, 26.49]	[24.0, 24.2]
q (median)	0.867	0.83
dL (Mpc)	~ 540	540^{+130}_{-210}
$\ln Z$	-104721.3 ± 0.23	N/A

Table 5. PE results vs published values. The $\sim 2 M_\odot$ chirp mass offset arises because this analysis does not apply the

cosmological redshift correction: $(1+z) \approx 1.12$ for $z \approx 0.12$, giving $26.2/1.12 \approx 23.4 M_\odot$, consistent with the published source-frame value of $24.1 M_\odot$. Mass ratio and distance are in excellent agreement.

6. Discussion

6.1 Value of Three-Detector Detection

GW170814 demonstrated three distinct advantages of the global network:

- Sky localisation. Inter-detector time delays and amplitude ratios between H1, L1, and V1 reduced the 90% sky area from $\sim 1160 \text{ deg}^2$ (two-detector) to $\sim 60 \text{ deg}^2$, enabling targeted electromagnetic follow-up.
- Polarisation test. With three non-co-planar baselines it is possible to project the measured strain onto the tensor, vector, and scalar polarisation modes predicted by GR and modified gravity theories. GW170814 provided the first such test, strongly favouring pure tensor polarisation.
- Artefact rejection. Coherent detection in three geographically separated detectors essentially rules out local instrumental transients as the signal origin.

6.2 Analysis Limitations

- Fixed sky/spin parameters. The published analysis marginalises over all 15+ parameters simultaneously; fixing sky position and spins reduces posterior volume but can bias estimates.
- No redshift correction. Detector-frame masses are $(1+z) \approx 1.12\times$ source-frame masses for $z \approx 0.12$.
- Template mismatch. The matched filter used GW150914 template masses ($36+29 M_\odot$), reducing SNR relative to the optimal GW170814 template ($\sim 30.5+25.3 M_\odot$).
- Limited live points. $N_{\text{live}} = 250$ gives approximate posteriors; publication quality requires ≥ 1000 .

6.3 GW150914 vs GW170814

Property	GW150914	GW170814
Detectors	H1 + L1	H1+L1+V1
$m_1 (M_\odot)$	36.2	30.5
$m_2 (M_\odot)$	29.1	25.3
$\mathcal{M}c (M_\odot)$	28.3	24.1
dL (Mpc)	~ 410	~ 540
90% sky (deg^2)	~ 230	~ 60
Network SNR	~ 24	~ 18

Table 6. Comparison of GW150914 and GW170814.

7. Conclusion

A complete GW analysis pipeline was applied to GW170814 — the first BBH merger observed coherently by three interferometers — using open GWOSC data. The three stages demonstrated:

- Preprocessing: ASD noise curves characterised the sensitivity of H1, L1, and V1. Q-transforms revealed

the GW170814 chirp sweeping from ~ 30 Hz to merger in all three detectors. The whitened strain showed the signal emerging coherently across the network.

- Matched filtering: Peak SNRs of 12.72 (H1), 12.53 (L1), and 5.14 (V1) were recovered using a BBH template. Virgo's lower SNR is fully explained by its higher noise floor during O2.
- Parameter estimation: The Dynesty nested sampler recovered detector-frame chirp mass $\mathcal{M}_c = 26.19 M_\odot$ (90% C.I.: [26.02, 26.49] $M_\odot$), $q = 0.87$, and $dL \approx 540$ Mpc, in excellent agreement with published LIGO/Virgo values after applying the redshift correction.

The pipeline is modular, fully documented, and extends naturally to other events, waveform models, and full-parameter sampling with larger N_{live} .

References

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